Year 10 Higher

REVISION PACK

Progress Check 2

ANSWERS

Units in this pack

- Graphs 1
- Number 2
- Shape and Space 1

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Q	Working	Answer
a missing coordinate		
1.1	$\frac{k-2}{7-3}=4$ $k-2=16$ $k=18$	$k=18$
1.2	$\frac{k-9}{4-1}=-2$ $k-9=-6$ $k=3$	$k=3$
1.3	$\frac{k-5}{3+2}=3$ $k-5=15$ $k=20$	$k=20$
1.4	$\frac{k--3}{4-0}=2$ $k--3=8$ $k=5$	$k=5$
1.5	$\frac{k-1}{5-2}=-3$ $k-1=-9$ $k=-8$	$k=-8$
1.6	$\frac{k-6}{3+1}=2$ $k-6=8$ $k=14$	$k=14$
the equation of a line through two points		
1.7	$m=\frac{12}{3}=4$ $7=4\times(2)+c$ $c=-1$	$y=4x-1$

1.8	$m = \frac{12}{4} = 3$ $-1 = 3 \times (-1) + c$ $c = 2$	$y = 3x + 2$
1.9	$m = \frac{-9}{3} = -3$ $7 = -3 \times (-1) + c$ $c = 4$	$y = -3x + 4$
1.10	$m = \frac{-12}{3} = -4$ $6 = -4 \times (1) + c$ $c = 10$	$y = -4x + 10$
1.11	$m = \frac{-8}{4} = -2$ $10 = -2 \times (-3) + c$ $c = 4$	$y = -2x + 4$
1.12	$m = \frac{10}{2} = 5$ $-4 = 5 \times (0) + c$ $c = -4$	$y = 5x - 4$

Number 2 · Percentages (Lesson 4.1)

Q	Working	Answer
percentage loss		
2.1	$18\,000 - 14\,940 = 3060$ $\frac{3060}{18\,000} \times 100 = 17\%$	**17%**
2.2	$640 - 528 = 112$ $\frac{112}{640} \times 100 = 17.5\%$	**17.5%**
2.3	$2500 - 2150 = 350$ $\frac{350}{2500} \times 100 = 14\%$	**14%**
2.4	$12\,000 - 9000 = 3000$ $\frac{3000}{12\,000} \times 100 = 25\%$	**25%**

2.5	$450 - 396 = 54$ $\frac{54}{450} \times 100 = 12\%$	**12%**
2.6	$8000 - 7400 = 600$ $\frac{600}{8000} \times 100 = 7.5\%$	**7.5%**
reverse percentages		
2.7	$100\% - 24\% = 76\%$ $5320 \div 0.76 = 7000$	**KD 7000**
2.8	$100\% - 15\% = 85\%$ $3060 \div 0.85 = 3600$	**KD 3600**
2.9	$100\% - 8\% = 92\%$ $391 \div 0.92 = 425$	**KD 425**
2.10	$100\% - 15\% = 85\%$ $204 \div 0.85 = 240$	**KD 240**
2.11	$100\% - 5\% = 95\%$ $2850 \div 0.95 = 3000$	**KD 3000**
2.12	$100\% - 20\% = 80\%$ $8320 \div 0.8 = 10\,400$	**KD 10 400**

Number 2 · Recurring decimals (Lesson 4.2)

Q	Working	Answer
recurring decimal proofs		
3.1	$x = 0.27777777\ldots$ $100x = 27.777777\ldots$ $10x = 2.777777\ldots$ $90x = 25$ $x = \frac{25}{90} = \frac{5}{18}$	$x = \frac{5}{18}$ **shown**

3.2	$x = 0.38888888\ldots$ $100x = 38.888888\ldots$ $10x = 3.888888\ldots$ $90x = 35$ $x = \frac{35}{90} = \frac{7}{18}$	$x = \frac{7}{18}$ **shown**
3.3	$x = 0.41111111\ldots$ $100x = 41.111111\ldots$ $10x = 4.111111\ldots$ $90x = 37$ $x = \frac{37}{90} = \frac{37}{90}$	$x = \frac{37}{90}$ **shown**
3.4	$x = 0.15555555\ldots$ $100x = 15.555555\ldots$ $10x = 1.555555\ldots$ $90x = 14$ $x = \frac{14}{90} = \frac{7}{45}$	$x = \frac{7}{45}$ **shown**
3.5	$x = 0.25555555\ldots$ $100x = 25.555555\ldots$ $10x = 2.555555\ldots$ $90x = 23$ $x = \frac{23}{90} = \frac{23}{90}$	$x = \frac{23}{90}$ **shown**
3.6	$x = 0.52222222\ldots$ $100x = 52.222222\ldots$ $10x = 5.222222\ldots$ $90x = 47$ $x = \frac{47}{90} = \frac{47}{90}$	$x = \frac{47}{90}$ **shown**

Shape and Space 1 · Angles (Lessons 5.1, 5.3)

Q	Working	Answer
the exterior angle of a triangle		

4.1	$x + 20 + 2x + 10 = 4x - 10$ $3x + 30 = 4x - 10$ $x = 40$ angle $ACD = 150^\circ$	**150°**
4.2	$2x + 15 + x + 20 = 4x - 10$ $3x + 35 = 4x - 10$ $x = 45$ angle $ACD = 170^\circ$	**170°**
4.3	$x + 15 + 3x - 5 = 5x - 20$ $4x + 10 = 5x - 20$ $x = 30$ angle $ACD = 130^\circ$	**130°**
4.4	$x + 30 + 2x + 5 = 4x - 5$ $3x + 35 = 4x - 5$ $x = 40$ angle $ACD = 155^\circ$	**155°**
4.5	$2x + 5 + x + 15 = 4x - 6$ $3x + 20 = 4x - 6$ $x = 26$ angle $ACD = 98^\circ$	**98°**
4.6	$x + 25 + x + 35 = 3x + 10$ $2x + 60 = 3x + 10$ $x = 50$ angle $ACD = 160^\circ$	**160°**
angles in a parallelogram		
4.7	$3x + 20 + 2x + 10 = 180$ (co-interior) $5x + 30 = 180$ $x = 30$ angles: 110° and 70°	**70°**

4.8	$3x - 15 + 2x + 5 = 180$ (co-interior) $5x - 10 = 180$ $x = 38$ angles: 99° and 81°	**81°**
4.9	$2x + 40 + 3x - 10 = 180$ (co-interior) $5x + 30 = 180$ $x = 30$ angles: 100° and 80°	**80°**
4.10	$4x + 10 + 2x + 20 = 180$ (co-interior) $6x + 30 = 180$ $x = 25$ angles: 110° and 70°	**70°**
4.11	$3x + 10 + 3x + 20 = 180$ (co-interior) $6x + 30 = 180$ $x = 25$ angles: 85° and 95°	**85°**
4.12	$2x + 25 + 4x + 5 = 180$ (co-interior) $6x + 30 = 180$ $x = 25$ angles: 75° and 105°	**75°**

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Q	Working	Answer
finding a shorter side		
5.1	$x^2 = 15.6^2 - 9.3^2 = 156.87$ $x = \sqrt{156.87} = 12.5248\ldots$	x **= 12.5 cm**
5.2	$x^2 = 11.8^2 - 6.4^2 = 98.28$ $x = \sqrt{98.28} = 9.91363\ldots$	x **= 9.91 cm**
5.3	$x^2 = 20.5^2 - 13.7^2 = 232.56$ $x = \sqrt{232.56} = 15.2499\ldots$	x **= 15.2 cm**

5.4	$x^2 = 9.4^2 - 5.1^2 = 62.35$ $x = \sqrt{62.35} = 7.8962\ldots$	x **= 7.9 cm**
5.5	$x^2 = 17.2^2 - 10.8^2 = 179.2$ $x = \sqrt{179.2} = 13.3866\ldots$	x **= 13.4 cm**
5.6	$x^2 = 13.5^2 - 8.9^2 = 103.04$ $x = \sqrt{103.04} = 10.1509\ldots$	x **= 10.2 cm**
the area of a trapezium		
5.7	$\frac{1}{2}(x + x + 3) \times 8 = 60$ $(2x + 3) \times 8 = 120$ $2x + 3 = 15$ $x = 6$	$x = 6$
5.8	$\frac{1}{2}(x + x + 4) \times 7 = 98$ $(2x + 4) \times 7 = 196$ $2x + 4 = 28$ $x = 12$	$x = 12$
5.9	$\frac{1}{2}(x + x + 6) \times 4 = 48$ $(2x + 6) \times 4 = 96$ $2x + 6 = 24$ $x = 9$	$x = 9$
5.10	$\frac{1}{2}(x + x + 6) \times 7 = 91$ $(2x + 6) \times 7 = 182$ $2x + 6 = 26$ $x = 10$	$x = 10$
5.11	$\frac{1}{2}(x + x + 4) \times 5 = 55$ $(2x + 4) \times 5 = 110$ $2x + 4 = 22$ $x = 9$	$x = 9$
5.12	$\frac{1}{2}(x + x + 2) \times 8 = 120$ $(2x + 2) \times 8 = 240$ $2x + 2 = 30$ $x = 14$	$x = 14$

radius from the area		
5.13	$\pi r^2 = 120$ $r^2 = 120 \div \pi = 38.1972\ldots$ $r = \sqrt{38.1972\ldots} = 6.18039\ldots$	**6.18 cm**
5.14	$\pi r^2 = 300$ $r^2 = 300 \div \pi = 95.493\ldots$ $r = \sqrt{95.493\ldots} = 9.77205\ldots$	**9.77 cm**
5.15	$\pi r^2 = 85$ $r^2 = 85 \div \pi = 27.0563\ldots$ $r = \sqrt{27.0563\ldots} = 5.20157\ldots$	**5.2 cm**
5.16	$\pi r^2 = 60$ $r^2 = 60 \div \pi = 19.0986\ldots$ $r = \sqrt{19.0986\ldots} = 4.37019\ldots$	**4.37 cm**
5.17	$\pi r^2 = 250$ $r^2 = 250 \div \pi = 79.5775\ldots$ $r = \sqrt{79.5775\ldots} = 8.92062\ldots$	**8.92 cm**
5.18	$\pi r^2 = 45$ $r^2 = 45 \div \pi = 14.3239\ldots$ $r = \sqrt{14.3239\ldots} = 3.7847\ldots$	**3.78 cm**